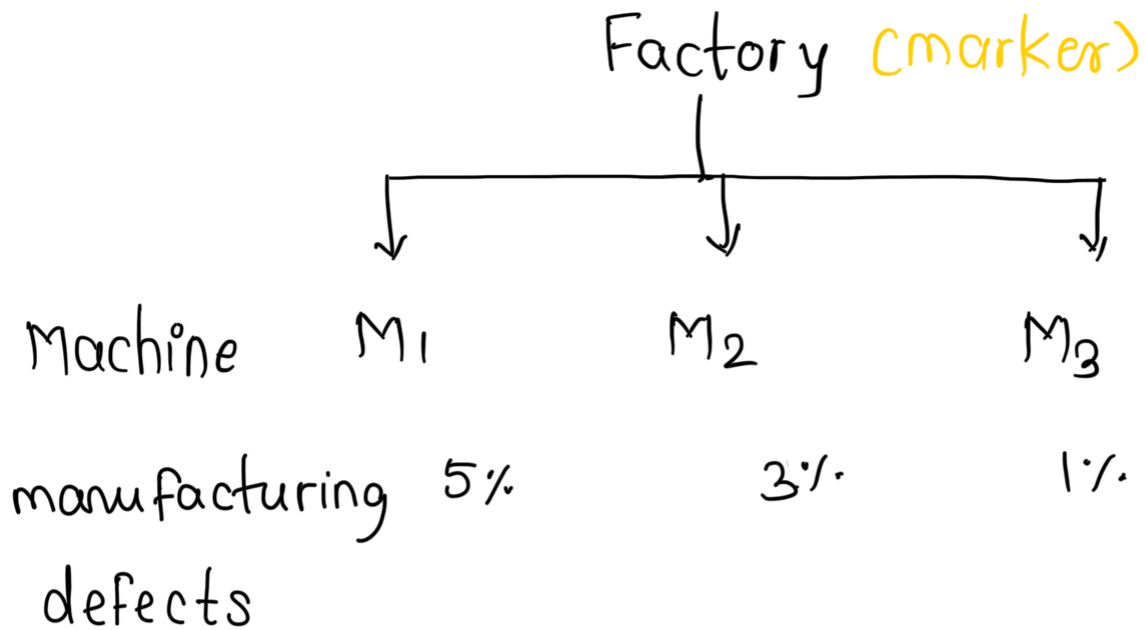


Bayes Theorem Examples



Question What is the probability of selecting a defective marker manufactured by M_3

$P(M_1) = \frac{20}{100} = \frac{1}{5}$ $P(M_2) = \frac{30}{100} = \frac{3}{10}$	D = Defective $P(D M_1) = \frac{5}{100} = \frac{1}{20}$ $P(D M_2) = \frac{3}{100}$
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$$P(M_3) = \frac{50}{100} = \frac{1}{2} \quad \Bigg| \quad P(D|M_3) = \frac{1}{100}$$

$$\therefore P(M_3|D) = \frac{P(D|M_3) \times P(M_3)}{P(D)}$$

$$P(D) = P(D \cap M_1) + P(D \cap M_2) + P(D \cap M_3)$$

$$P(A|B) = \frac{P(A \cap B)}{P(B)} \quad \therefore P(A \cap B) = P(A|B) \times P(B)$$

$$\therefore P(D) = P(D|M_1) \times P(M_1) + P(D|M_2) \times P(M_2) + P(D|M_3) \times P(M_3)$$

$$= \frac{1}{20} \times \frac{1}{5} + \frac{3}{100} \times \frac{30}{100} + \frac{1}{100} \times \frac{1}{2}$$

$$= \underline{\underline{\frac{1}{100}}} + \underline{\underline{3}} \times \underline{\underline{30}} + \underline{\underline{\frac{1}{200}}}$$

$$\overline{100} \quad \overline{100} \quad \overline{100} \quad \overline{100} \quad \overline{2}$$

$$= \frac{1}{100} \left[1 + 3 \times 30 + \frac{1}{2} \right]$$

$$= \frac{1}{100} [1 + 90 + 0.5]$$

$$= \frac{91.5}{100}$$

$$\therefore P(m_3 | D) = \frac{1}{\cancel{100}} \times \frac{\cancel{50}}{\cancel{100}} \times \frac{\cancel{100}}{91.5}$$

$$P(m_3 | D) = \frac{5}{915}$$